

Breast Cancer Tumor Classification



Using Clinical Features & Machine Learning

Logistic Regression

SVM (RBF Kernel)

Random Forest

SHAP Explainability



Problem Statement

The Challenge

Accurate classification of breast tumors as Malignant or Benign from digitized cell nuclei images.

Clinical Impact

Early detection of malignant tumors is critical; it directly improves patient survival rates and guides treatment decisions.

Objective

Build & compare ML models achieving >95% accuracy, with high recall to minimize missed cancer cases (False Negatives).

Dataset at a Glance

568

Total Samples

30

Feature Columns

357

Benign (63%)

211

Malignant (37%)

0

Missing Values



Dataset Overview

Mean Values

- radius_mean
- texture_mean
- perimeter_mean
- area_mean
- smoothness_mean
- compactness_mean

Standard Error (SE)

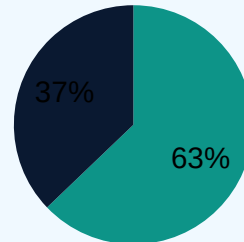
- radius_se
- texture_se
- perimeter_se
- area_se
- smoothness_se
- compactness_se

Worst Values

- radius_worst
- texture_worst
- perimeter_worst
- area_worst
- concave_points_worst
- concavity_worst

Target Variable: outcome

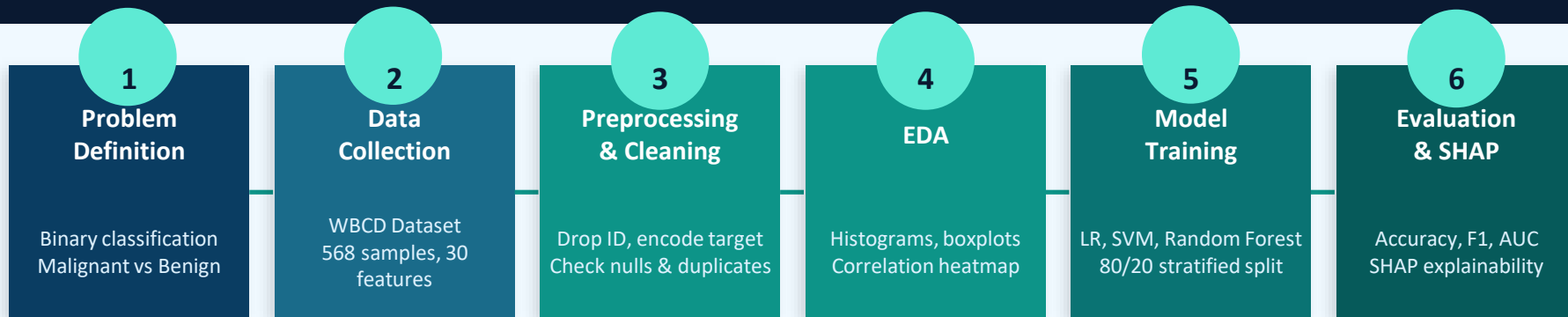
M = Malignant (1) | B = Benign (0)



- Benign (63%)
- Malignant (37%)



Methodology & Pipeline



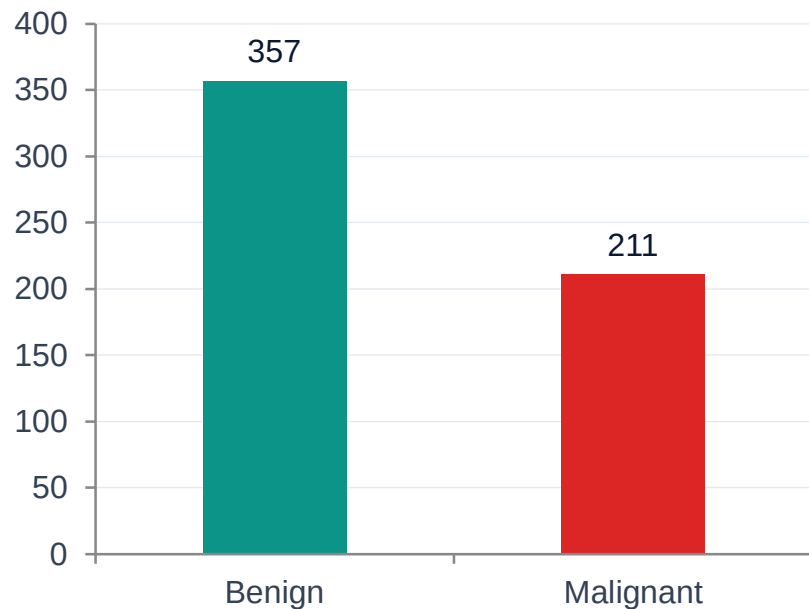
Key Techniques Applied

- ✓ StandardScaler for feature normalization
- ✓ Stratified 80/20 train-test split
- ✓ 5-Fold Stratified Cross-Validation
- ✓ SHAP & Permutation Importance for model explainability
- ✓ Confusion Matrices & ROC-AUC for evaluation
- ✓ joblib model serialization (.pkl files)



Exploratory Data Analysis

Class Distribution



Key EDA Findings

- Class Balance:**
Dataset is slightly imbalanced; 63% Benign vs 37% Malignant. Suitable for stratified splitting.
- Feature Distributions:**
Most features show normal/skewed distributions. Histograms reveal overlap between classes.
- Outliers Detected:**
Boxplots reveal outliers in several features; not extreme enough to impact analysis significantly.
- High Correlations:**
Radius, Perimeter & Area are strongly correlated (redundancy). Concave features are key discriminators.
- Clear Separation:**
Scatter plot of radius_mean vs area_mean shows clear visual separation between Malignant & Benign.

Model Building & Training

Logistic Regression

Parameters

```
max_iter=1000  
random_state=42
```

Advantages

- ✓ Simple & interpretable
- ✓ Fast training
- ✓ Good baseline

SVM (RBF Kernel)

Parameters

```
kernel='rbf'  
probability=True
```

Advantages

- ✓ Handles non-linearity
- ✓ Robust to outliers
- ✓ Strong generalization

Random Forest

Parameters

```
n_estimators=100  
random_state=42
```

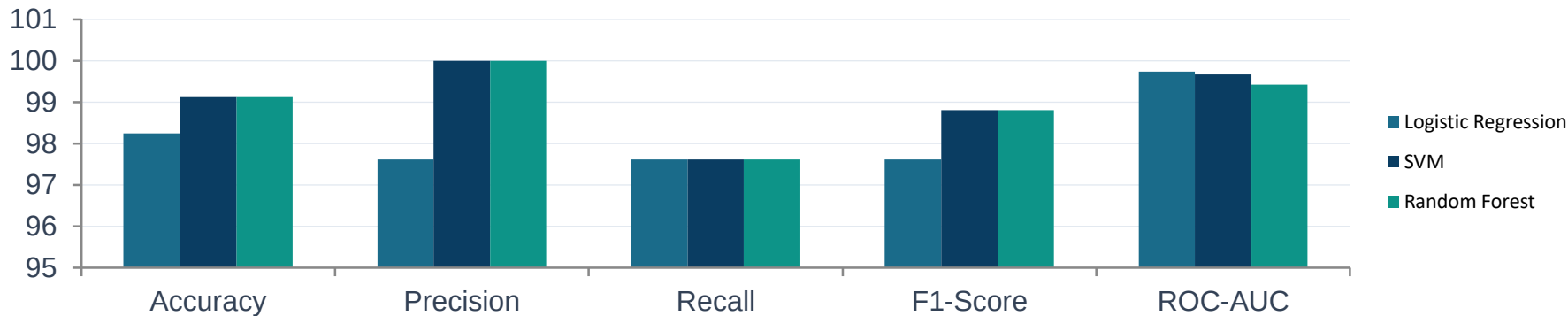
Advantages

- ✓ Ensemble method
- ✓ Feature importance
- ✓ Handles redundancy



Key Results & Model Comparison

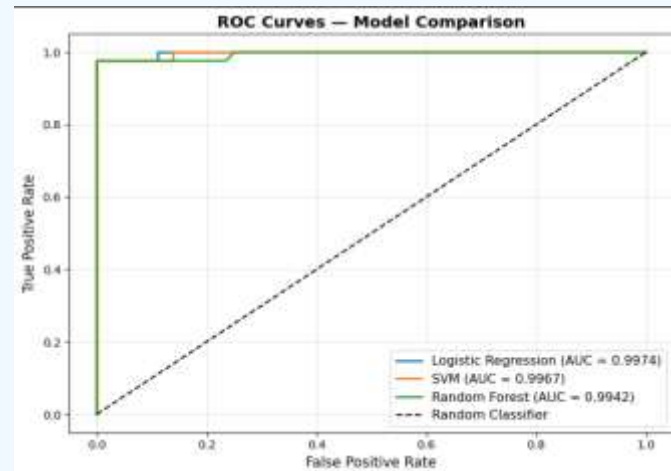
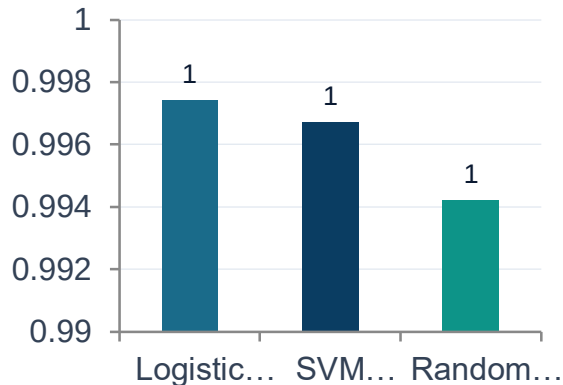
Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	98.25%	97.62%	97.62%	97.62%	99.74%
SVM (RBF Kernel)	99.12%	100.00%	97.62%	98.80%	99.67%
Random Forest	99.12%	100.00%	97.62%	98.80%	99.42%



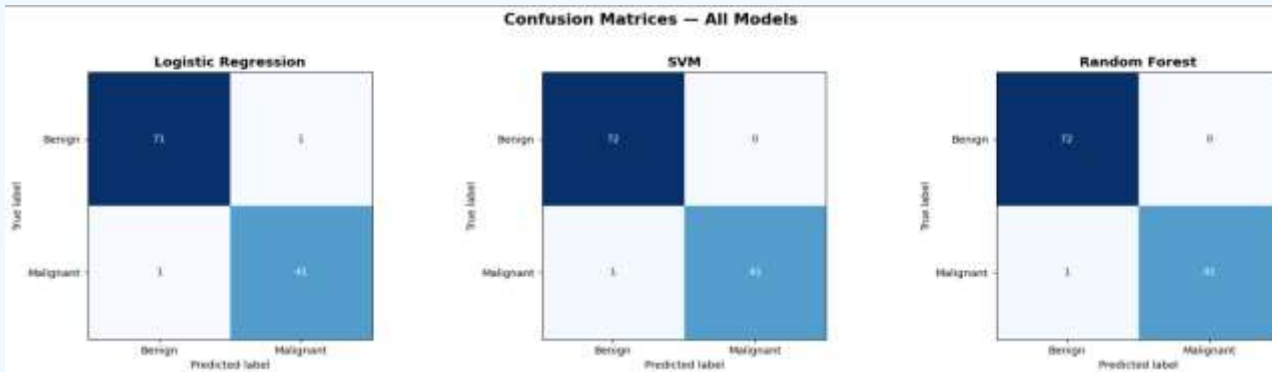
All three models exceed 98% accuracy — SVM & Random Forest tie at 99.12%

ROC Curves & Evaluation Charts

ROC-AUC Comparison



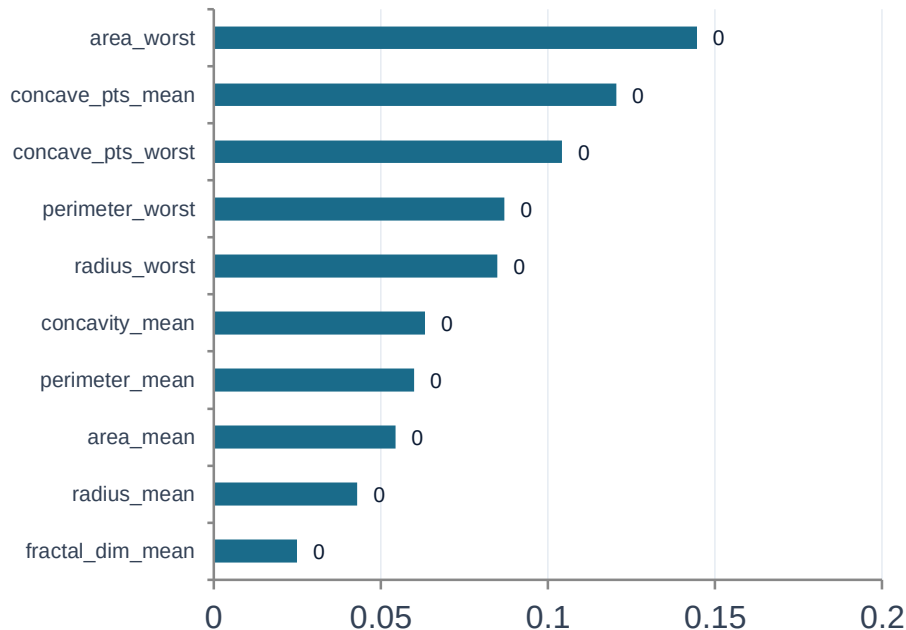
Confusion Matrices — All Models





Feature Importance & SHAP Explainability

Random Forest-Top 10 Features



SHAP Explainability Methods

RF Feature Importance

Global feature ranking by Gini impurity

LR Coefficients

Direction & magnitude of each feature's influence

Permutation Importance

True impact on F1 score per model

SHAP Summary (Dot)

Feature impact distribution across samples

SHAP Summary (Bar)

Mean absolute SHAP values (global importance)

SHAP Waterfall

Single prediction explained feature-by-feature

Key Finding: area_worst, concave_points_mean & concave_points_worst are the most consistent predictors across all models.



Deployed Application

A screenshot of a web application window titled "Streamlit App". It features a teal header bar with three colored window control buttons (red, orange, green) and a title bar. Below the header is a dark blue main area containing a teal horizontal bar at the top, followed by five white horizontal input fields. At the bottom of the main area is a teal button labeled "Predict".

Input Form

User enters 30 clinical feature values via an interactive web form built with Streamlit.

A screenshot of the same web application window showing the prediction output. The teal header bar and five white input fields are visible. The "Predict" button is now a teal button labeled "Predict". Below the button is a large red rectangular box containing the text "⚠ MALIGNANT" and "95.3% confidence" in white.

Prediction Output

Model returns classification result (Malignant/Benign) with probability confidence score.

A screenshot of the same web application window showing the model selection screen. The teal header bar and five white input fields are visible. The "Predict" button is now a teal button labeled "Predict". Below the button is a large red rectangular box containing the text "⚠ MALIGNANT" and "95.3% confidence" in white.

Model Selection

Users can select between Logistic Regression, SVM, or Random Forest for prediction.

Conclusions, Limitations & Future Work



Conclusions

- ▶ All 3 models achieved >98% accuracy on the Wisconsin Breast Cancer Dataset.
- ▶ SVM & Random Forest tied at 99.12% accuracy with 100% Precision.
- ▶ Logistic Regression achieved the highest ROC-AUC (0.9974); best for screening.
- ▶ Key features: area_worst, concave_points_mean, perimeter_worst.
- ▶ SHAP analysis confirms consistent feature importance across all models.



Limitations

- ▶ Dataset is relatively small (568 samples); limited generalizability.
- ▶ Slight class imbalance (63% Benign vs 37% Malignant) may affect minority class.
- ▶ Features are pre-computed from images; raw image data not used.
- ▶ Models not validated on external clinical datasets.
- ▶ No hyperparameter tuning performed (Grid Search / Bayesian Optimization).



Future Work

- ▶ Apply deep learning (CNN) directly on cell nuclei images.
- ▶ Perform hyperparameter optimization with GridSearchCV.
- ▶ Handle class imbalance using SMOTE or class weights.
- ▶ Validate on real-world hospital datasets for clinical applicability.
- ▶ Integrate SHAP explanations into the deployed Streamlit app.